MTCSAR: Fusing Multitemporal Interaction With Coherence Prior for SAR Image Denoising

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***Abstract*— Synthetic aperture radar (SAR) images are inherently affected by speckle noise due to their imaging principles, significantly impacting downstream research. Denoising methods based on deep learning have garnered attention and are gradually maturing, yet they face certain challenges. Denoising single SAR images lacks temporal information, often resulting in structural fitting that introduces artifacts. The existing denoising techniques do not adequately account for the differences between SAR and optical images, despite their distinct structural characteristics. In addition, mainstream deep learning approaches heavily rely on data-driven methods, with limited consideration for statistical properties. In response to these challenges, we propose a novel network for denoising SAR images fusing multitemporal interaction with coherence prior, termed MTCSAR. The proposed network leverages multitemporal interaction (MTI) to gather information from different moments in various regions. The redundancy across these temporal dimensions helps mitigate artifacts and edge blurring. To address SAR image characteristics, we design a dual-branch network combining global and local information to handle large-scale regions and fine structures. In addition, we incorporate prior coherence information from SAR images into the network, utilizing statistical properties to enhance transparency during training. Experimental results on simulated and real datasets demonstrate that injecting MTIs and coherence improves our method's qualitative and quantitative performance, surpassing current state-of-the-art algorithms. This validates the effectiveness of the proposed MTCSAR for multitemporal SAR image denoising.**

***Index Terms*— Deep learning, image despeckling, multitemporal synthetic aperture radar (SAR), physical prior.**

I. Introduction

A SYNTHETIC aperture radar (SAR) is an active remote sensing technique for ground observation, capable of acquiring data under all weather conditions and at any time. Unlike visible light and infrared images, SAR can penetrate soil and vegetation more effectively, allowing for the retrieval of surface vegetation and soil properties. The sensor receives polarized data from the returned signals, providing information about surface structures and moisture content. Consequently, SAR is widely used in urban monitoring, land classification, vegetation assessment, and deformation studies [1], [2], [3], [4]. However, in practical applications, the measured signals are affected by speckle noise. This noise arises from SAR's inherent characteristics: it forms images by transmitting microwave signals and receiving backscattered echoes. Irregular surface scatterers cause variations in the coherence of the returned signals, resulting in speckle patterns on the images. Removing this noise is crucial to mitigate interference in subsequent SAR applications and restore clean underlying information [5], [6], [7].

The noise generated during microwave scattering from ground objects cannot be eliminated through hardware upgrades. SAR image denoising techniques operate at the algorithmic level to restore ground features and mitigate clutter effects. Due to significant differences between SAR image noise and optical image noise, specialized denoising algorithms are necessary. Early approaches relied on spatial filtering based solely on pixel grayscale correlations within the image [8], [9], [10], [11]. These methods used neighborhood or sliding windows weighted computations to estimate pixel grayscale values within the target window. However, they often result in pronounced smoothing effects, which are effective in uniform regions but cause blurring and loss of detail at edges. To address this, researchers incorporated transform domain information into denoising models. For instance, the block-matching and 3-D filtering (BM3D) algorithm combined fast matching and 3-D transform domains [12], partially addressing the issue of information loss in denoised images. Subsequently, it was introduced to the SAR denoising [13], utilizing a local linear minimum mean square error filter. Nevertheless, due to the limitations of single-image information, denoised results from individual temporal images can only approximate noise-affected pixels based on surrounding structures, potentially leading to unsatisfactory results.

The introduction of multitemporal data can address these challenges by providing additional temporal information, thereby enhancing algorithm robustness. The two-step PPB nonlocal filter (2S-PPB) method [14] extends probability patch-based (PPB) [15] filtering to multitemporal SAR images, naturally adapting [11] to multitemporal tasks. It employs a two-step strategy for spatial nonlocal weighted estimation driven by temporal redundancy. Similarly, the multitemporal SAR-oriented version of BM3D (MSAR-BM3D) [16] approach extends SAR-BM3D [13] to handle multitemporal

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data using two strategies, relying on nonlocal methods based on block matching and collaborative filtering. Other methods leverage multitemporal features. Quegan et al. [17] use time-weighted averaging with linear multichannel intensity filters, while Coltuc et al. [18] incorporate weighted averaging in the transform domain, applying discrete cosine transform along the time axis and adaptive filters on the transformed results. Ratio-based multitemporal SAR denoising framework (RABASAR) [19] computes a "super-image" from the stack of multilooks, providing ratio images of the target image, improving image structure and feature stability while preserving temporally thin structures. Despite these advancements, model-based algorithms still face limitations, as they cannot autonomously extract data features, leading to potential oversmoothing or artifacts, and they remain subject to rigorous optimization processes.

With the development of deep learning [16], [20], [21], [22], methods based on convolutional neural network (CNN) have gained widespread use in the field of denoising. These methods enable networks to learn data processing in an end-to-end manner [23], [24], [25], enhancing their versatility. Combining deep learning with multitemporal SAR images can address image artifacts and optimization challenges to some extent. The MSAR-CNN algorithm [26], which embeds spatiotemporal attention modules, leverages CNN to explore temporal correlations in multitemporal SAR images. In addition, self-supervised training methods have been employed, using real data to learn how to restore clean images solely from noise-corrupted ones [27]. SAR-to-SAR denoising method (SAR2SAR) [28] compensates for temporal changes and adapts to the speckle statistics of SAR images by introducing a loss function based on multitemporal time series. Ma et al. [29] utilize parameter-sharing CNNs, combining weighted losses through self-supervised training with complementary images from different times. Meraoumia et al. [30] extend the self-supervised training strategy MERLIN from single-look complex images to multitemporal filtering, modeling statistical relationships between the real and imaginary parts of complex amplitudes across both spatial and temporal dimensions. However, noise-reference-based self-supervised methods heavily rely on statistical assumptions and prior knowledge, which may lead to image blurring or artifacts [31]. These approaches often prioritize data-driven aspects and may not fully consider the statistical properties of multitemporal SAR images. Furthermore, CNN-based methods may exhibit weaknesses in capturing global dependencies, which is crucial for effective SAR image denoising.

Due to the strong global dependencies exhibited by Transformers [32], they have gradually found applications in image processing tasks [33], [34], [35], [36]. Xiao et al. [37] proposed the global context attention and top-k token selective Transformer to enhance traditional global attention mechanisms, achieving performance superior to both CNNs and standard Transformers. Inspired by video remote sensing super-resolution work [38], we recognize that local features are prominent at the boundaries of objects, while global features contain additional edge and shape details. The unique structure of SAR images underscores the importance of both global and local information in denoising tasks. To address this, we adopt a specially designed dual-branch strategy to fuse local and global information systematically. Leveraging the prior knowledge of SAR statistical properties allows the network to avoid irrelevant factors, restrict feature space, and guide training to learn the essence of the task. Furthermore, the existing multitemporal SAR image networks rarely incorporate physical prior information. To address this gap, we introduce coherence priors into the network, enhancing accuracy and improving the model's ability to capture critical information.

Specifically, we propose the multitemporal interaction with coherence prior network (MTCSAR) for SAR image denoising. We use coherence as a prior statistical characteristic and introduce it to multitemporal denoising tasks. The proposed multitemporal interaction (MTI) extracts global spatiotemporal features from long-term series, while local information is obtained from the coherence coefficient map (CCM) feature extraction (CFE) module in conjunction with the coherence prior. Our network cleverly complements this information. To alleviate potential misalignment caused by the fusion of these two types of information, the dual attention fusion (DAF) module enhances the interaction between target and neighboring images, adaptively learning the relationship between spatiotemporal features and CCM prior features. In this way, our MTI and CFE modules can concentrate on more valuable information to reconstruct the target, maintaining spatiotemporal consistency.

The contributions of this article are as follows.

1) We propose a network for multitemporal SAR image denoising, which can capture the global information of SAR images and the local information of CCM. The two complement each other to promote denoising.
2) Our MTI module extends spatial self-attention to the spatiotemporal dimension, capturing a broader context. This approach overcomes the limitations of CNN methods that focus solely on local spatial information.
3) We construct a DAF architecture that utilizes independent temporal attention and prior pixel attention to effectively extract crucial spatiotemporal information from multitemporal SAR images.

The remainder of this article is organized as follows. Section II reviews SAR-related theory. Section III details our proposed method. Section IV presents experiments and analysis. Section V discusses the extension and implications of the proposed method, and Section VI concludes this article.

II. Principle of Related Theories

A. Speckle Model

According to the speckle model proposed in [39], observed SAR signal intensity Y can be expressed as follows:

$$Y = X \cdot N \tag{1}$$

where X is true signal without interference and N is the speckle noise. Nikolova [40] assumes that this multiplicative noise follows gamma distribution, which is

$$P(n) = \frac{1}{\Gamma(n)} L^L n^{L-1} e^{-Ln} \tag{2}$$

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where Γ is gamma function and $L \geq 1$ denotes the number of looks. The mean of this distribution is 1, and the variance is reciprocal of L.

The denoising task can be expressed as follows:

$$\hat{X} = f(Y, \Theta). \tag{3}$$

Its goal is to obtain a parameter set Θ that enables the network output $\hat{X}$ to closely fit the reference X, minimizing the loss between them (4) to reconstruct the mapping from Y to X. It is

$$\arg\min_{\Theta} \text{Loss}\left[\hat{X}, X\right]. \tag{4}$$

By adjusting L, we can simulate different intensities of speckle noise, providing the network with diverse training data.

B. Coherence and Coherence Coefficient Map

Coherence is a physical property in multi-SAR images [41] commonly used in interferometric SAR [42]. It quantifies the similarity between multiple complex images. Coherence is defined as follows:

$$\gamma = \frac{\left|E\left[S_1 S_2^*\right]\right|}{\sqrt{E\left[|S_1|^2\right] E\left[|S_2|^2\right]}} \tag{5}$$

where S_1 and S_2 represent the complex values of two pixels in temporally separated SAR images, $*$ represents conjugation, E denotes the mathematical expectation, and γ signifies the coherence coefficient.

Regions where two images exhibit higher similarity correspond to stronger coherence. By applying (5), we can compute the similarity between adjacent temporal SAR images relative to the reference temporal image. The coherence coefficient reflects the reliability of information captured in the temporal image and ranges from 0 to 1. The closer the coefficient is to 1, the higher the similarity between images. For each pixel in the map, the coherence coefficient is calculated using (5) and used as the value of that pixel. These pixels together form the CCM.

CCM has been widely applied in various fields, such as flood monitoring, building time-series tracking, and vegetation research [43], [44], [45], [46]. In the context of multitemporal image denoising, the CCM can provide prior information about the availability of specific regions within the target image across all time stacks, enabling a better understanding of the terrain geometry within the scene.

III. Methodology

The proposed multitemporal SAR image denoising model in this article is illustrated in Fig. 1. It takes a stack of noisy multitemporal images and the corresponding CCM as input, to produce a high-quality denoised image of the target date. The model comprises several key components, including the MTI module, the CFE module, and the DAF module.

Assuming the model receives multitemporal SAR images with noise, denoted as Y_M, and their corresponding CCM, denoted as CC_{M-1}, where M represents the number of temporal images. MTI extracts spatiotemporal global features from Y_M, that is,

$$X_{\text{global}} = H_{\text{MTI}}(\text{Conv}(Y_M)) \tag{6}$$

where H_{MTI} denotes the MTI module and X_{global} means the extracted global spatiotemporal features.

At the same time, the CCM calculated from Y_M passes through CFE module to extract shallow prior local features. It is shown as follows:

$$\text{CC}_{\text{local}} = H_{\text{CFE}}(\text{CC}_{M-1}) \tag{7}$$

where H_{CFE} is the function of CFE and CC_{local} is local information obtained from CC_{M-1}.

Afterward, X_{global} and CC_{local} are fused through a designed dual attention module DAF to obtain features enhanced by learnable parameter weights for time-adaptive and physical prior enhancement. The subsequent expression is

$$X_{\text{TA}} = H_{\text{TA}}\left(X_{\text{global}}\right) \tag{8}$$

$$X_{\text{CCMA}} = H_{\text{CCMA}}\left(X_{\text{global}}, \text{CC}_{\text{local}}\right) \tag{9}$$

$$X_{\text{fusion}} = \lambda_1 X_{\text{TA}} + \lambda_2 X_{\text{CCMA}}. \tag{10}$$

In (8), H_{TA} means adaptive temporal attention block. X_{TA} is the output of H_{TA}. In (9), H_{CCMA} denotes CCM attention block, and X_{CCMA} is the output of H_{CCMA}. In (10), X_{fusion} denotes fusion features after DAF processing, which contains both image itself and prior information.

Finally, the feature map is processed by a single convolutional layer to obtain the output noise mapping. Through global residual learning, the denoised image Y corresponding to the target temporal is obtained, which is

$$\hat{X} = \text{Conv}(X_{\text{fusion}}) + Y_{M_{\text{target}}} \tag{11}$$

where $\hat{X}$ is the denoised SAR image and $Y_{M_{\text{target}}}$ means the noise image of target reference.

In Sections III-A–III-D, we will delve into specific details of the main modules in MTCSAR.

A. Multitemporal Interaction

In the context of single-image feature extraction, self-attention mechanisms have gained popularity due to their high modeling capacity and excellent performance. For large-scale image processing, window-based methods [47] have also been employed to reduce computational complexity. When comparing multitemporal SAR images with single ones, a significant advantage lies in the substantial redundancy of information for the same area across different time points. To maximize the utilization of this advantage, we propose a novel approach called spatiotemporal anchor self-attention (SAS). SAS is specifically designed to handle the additional dimensions introduced by multitemporal images and their associated computational demands. Within our proposed framework, SAS, channel attention, and shortcut connections jointly form a basic unit known as the MTI layer (MTIL), as illustrated in Fig. 2. Multiple MTILs are then combined to create the MTI block (MTIB), responsible for extracting spatiotemporal

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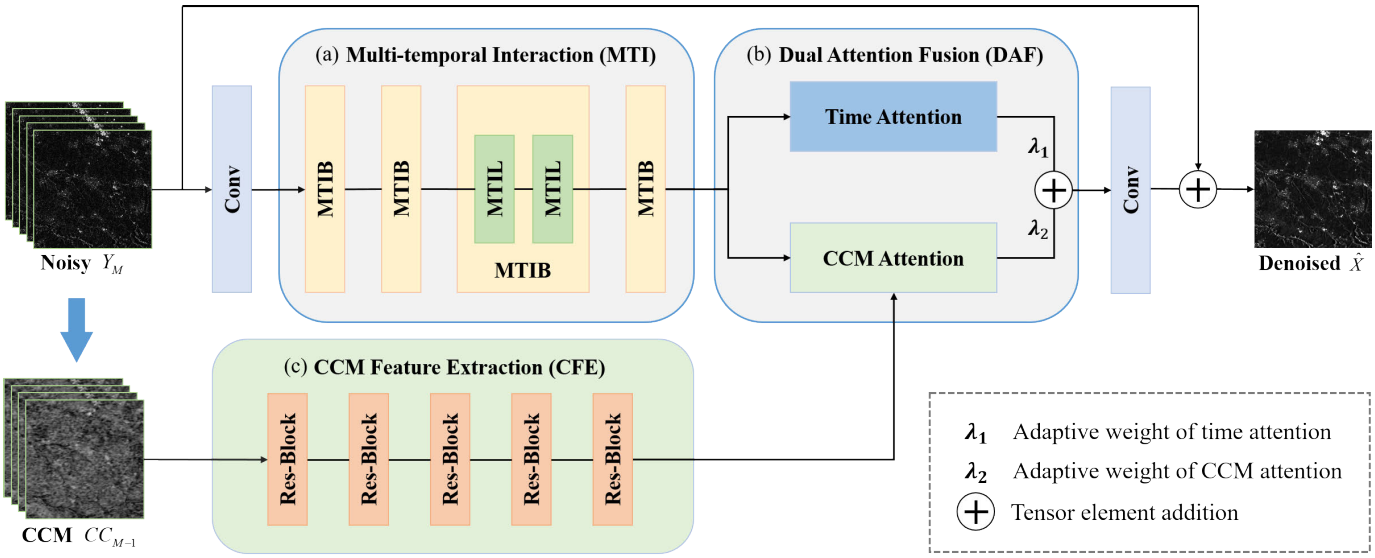


Fig. 1. Overall network architecture of MTCSAR. (a) MTI module contains several MTIBs. (b) DAF module is a dual-branch subnetwork. (c) CCM helps with image feature reconstruction.

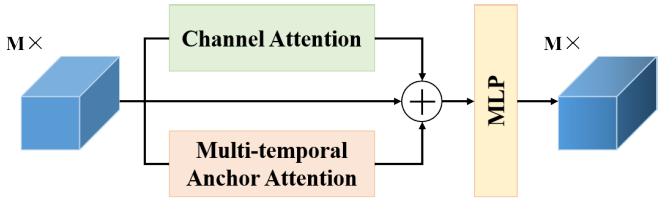


Fig. 2. MTIL.

information from images. The collective action of several MTIBs constitutes the MTI module.

In MTIL, we compute SAS and channel attention in parallel. Channel attention adaptively enhances the network's sensitivity to important channel information, allowing it to better recognize critical features. Let X_0 be the input to the MTIL. We process X_0 and subsequently use a multilayer perceptron (MLP) to delve into the contextual information of the features. The process can be represented as follows:

$$\text{Out}_{\text{MTIL}} = \text{MLP}(H_{\text{CA}}(X_0) + H_{\text{SAS}}(X_0) + X_0) \tag{12}$$

where H_{CA} means channel attention module and H_{SAS} is SAS.

We focus on the SAS module. While large-scale images introduce more redundant information, they also pose significant computational challenges [48], [49]. Even when employing the shifted window approach, the temporal dimension of multitemporal SAR images results in large windows, leading to high-computational complexity for self-attention calculations within these windows. Therefore, we introduce the concept of anchors [50] into multitemporal image processing to enhance efficiency. The multitemporal anchor serves as a summary of information from the image feature maps, acting as an intermediary for self-attention computations. It has been demonstrated that the attention maps using anchors are nearly identical to those of regular self-attention maps.

In general [32], self-attention calculations are represented by (13). After introducing the anchor, the self-attention calculation can be reformulated as (14). They are

$$X = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{13}$$

$$X = \text{softmax}\left(\frac{QA^T}{\sqrt{d_k}}\right)\cdot\left[\text{softmax}\left(\frac{AK^T}{\sqrt{d_k}}\right)\cdot V\right] \tag{14}$$

where X, Q, K, and $V \in R^{N\times d_k}$. N denote the number of tokens, and d_k is the dimension of token. $A \in R^{G\times d_k}$ represents the previously mentioned anchor, where $G \ll N$.

Consequently, the computational complexity of self-attention changes from $O(N^2 d_k)$ to $O(NGd_k)$. The image patch has dimensions $H \times W \times M$, while the window has dimensions $H_w \times W_w \times M$, with an additional temporal dimension M. After embedding, N_w windows are obtained within a single patch. Anchors serve as summaries of window information and are typically implemented as average pooling layers with a size of $H_w/s \times W_w/s \times M$, where s represents the level of summarization relative to the original feature map. The anchors participate in parallel computation along two paths, and the calculation process and intermediate results are illustrated in Fig. 3.

B. CCM Feature Extraction

CCM can capture differential change information and, together with global features, represent statistical characteristics in multitemporal SAR images. Its characteristics and physical meaning are provided in Section II-B. CCM utilizes complex image calculations, which contain richer and more accurate information compared with traditional log-ratio parameters. In this study, CCM computation is performed using SARscape, employing an adaptive window size to obtain results unaffected by noise. SARscape is a professional software for radar image processing, widely used in the field of SAR data analysis. It is typically integrated into the ENVI remote sensing image processing software and provides a graphical user interface [51].

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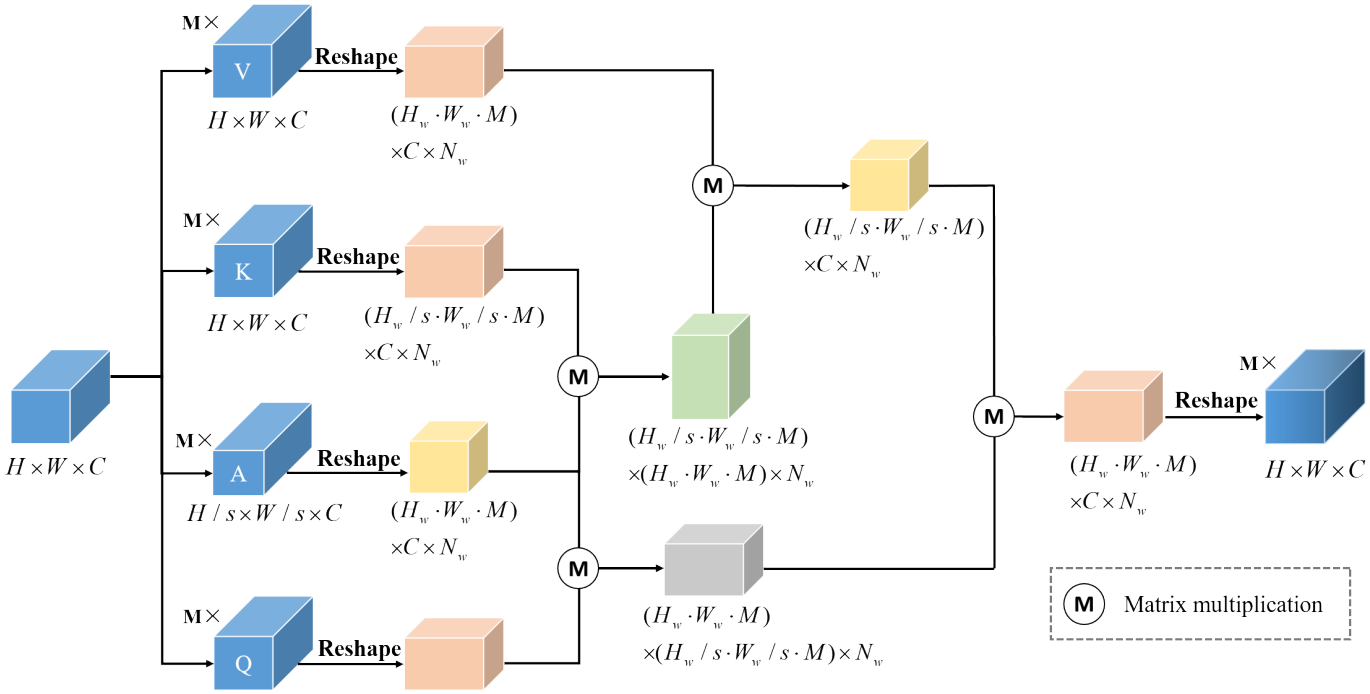


Fig. 3. Structure of SAS. The multitemporal anchor reduces the amount of calculation.

We use CCM as prior physical information. The core structure of this module consists of residual blocks [52], employing a parameter-sharing strategy to extract CCM features from temporally adjacent images relative to the reference image. Specifically, these CCMs are sequentially input into the module, resulting in the extraction of their corresponding feature maps as output. When the number of temporal images is M, coherence is calculated as the similarity between other images relative to the target image. Therefore, the number of CCMs should be $M - 1$.

The processing in residual dense block can be shown as follows:

$$X_i = \text{Conv}(\text{ReLU}(\text{Conv}(X_{i-1}))) + X_{i-1} \tag{15}$$

where X_{i-1} and X_i are the input and output of residual dense block.

C. Dual Attention Fusion

The local prior features of CCM can assist in denoising multitemporal SAR images, but an appropriate fusion method is required to combine local and global information. Considering the high temporal redundancy in multitemporal images, Wang et al. [53] utilize adaptive spatiotemporal attention to integrate this multitemporal information. To further fuse the spatiotemporal information, we employ temporal attention to process global features of these SAR images, enhancing the target temporal features. Simultaneously, the designed CCM attention module introduces local prior features of CCM into another branch of DAF. The results are weighted fusion through adaptively learned parameters, allowing the network to autonomously assess and balance the importance of both components (10), as shown in Fig. 1(b). This approach

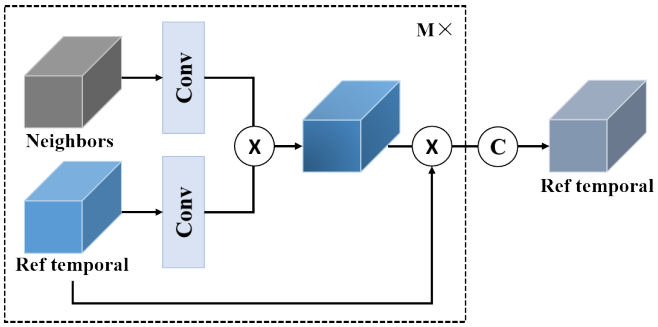


Fig. 4. Time attention between reference temporal image and neighbors. M means the length of the time series.

strengthens the network's ability to model temporal relationships, achieving precise information fusion at the feature level.

Specifically, for the temporal attention, as shown in Fig. 4, it primarily computes the similarity between each temporal image and the target image at the feature level. The similarity is measured by taking the elementwise product of each image. Normalization functions are then applied to obtain adaptive weights in the temporal domain. The detailed process is described in Algorithm 1.

In the fusion of prior information features, CCM features have already been extracted in Section III-B. Considering the inherent physical characteristics of CCM, we utilize the extracted features as weights to combine them with features from multitemporal images. This fusion occurs at the feature level. As shown in Fig. 5, if the inputs of this module are X_{global} and CC_{local}, then the output can be expressed as follows:

$$X_{\text{fusion cc}} = \text{Conv}\left(\text{CC}_{\text{local}} \cdot \text{Conv}\left(X_{\text{global}}\right)\right) \tag{16}$$

where $X_{\text{fusion cc}}$ is the output of CCM attention module.

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Algorithm 1 Processing Flow of Time Attention

Suppose M is the number of temporal images and j is the target

Input: Reference temporal feature F_j; Multi-temporal feature sequence F

Output: Processed reference temporal feature $\hat{F}_j$

for i = 1: M

$h(i, j) = \mathbf{sigmoid}(F_i{}^T \cdot F_j)$

$\widehat{F_{ij}} = h(i, j) \cdot F_j$

end

$\hat{F}_j = \mathbf{concat}(\widehat{F_{1:M,j}})$

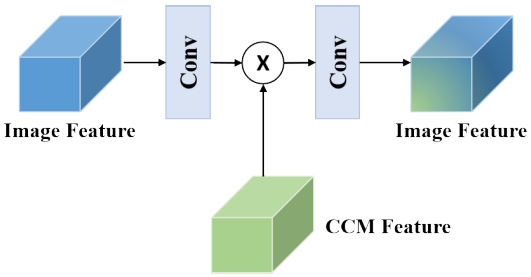


Fig. 5. CCM attention module. Training is guided by CCM information.

D. Loss Function

Compared with the commonly used mean-squared error (mse), Charbonnier loss [54] is more robust in handling outliers and exhibits greater smoothness in regions with high gradients. This property helps preserve fine details in SAR images. Consequently, we employ Charbonnier loss as the loss function during network training. Even in the presence of strong gamma noise interference, it enables the model to produce denoised SAR images that are clearer and more natural. The loss is expressed as follows:

$$L = \sqrt{||x - \hat{x}||^2 + \epsilon^2} \tag{17}$$

where $\hat{x}$ is denoised SAR image and ϵ is a very small number.

IV. Experiments

To validate the effectiveness, we compared the proposed method with nine commonly used image restoration techniques, including four model-based methods (unbiased temporal average (UTA) [17], MSAR-BM3D [16], RABASAR [19], and 2S-PPB [14]) and five data-driven algorithms (SAR2SAR [28], noisy reference-based SAR deep learning filter (NR-SAR) [29], MSAR-CNN [26], SAR despeckling using overcomplete convolutional networks (SAR-OCN) [55], and multi-objective network (MONet) [56]). It should be noted that there are very few multitemporal SAR image denoising networks; among deep learning methods, only SAR2SAR, NR-SAR, and MSAR-CNN are models used to process multitemporal SAR data. NR-SAR and SAR2SAR are trained with two temporal images, whereas MSAR-CNN utilizes five. Among the model-based approaches, MSAR-BM3D processes feature from four temporal images, while the other methods also use five. To ensure experimental fairness, the proposed network was trained and tested on data from two, four, and five temporal images, making the comparison with other models more convincing. The parameters for model-based algorithms were adjusted according to their respective papers, while the data-driven methods were trained on the dataset used in this study. To explore the significance of the proposed modules in the network, we conducted ablation experiments by comparing the performance of networks with specific modules removed against the complete network. All tests were conducted on a PC equipped with four NVIDIA RTX 3090 Ti GPUs and 128 GB of memory.

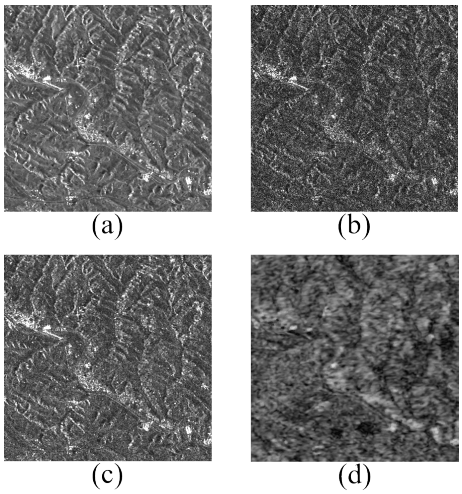


Fig. 6. Subset of our dataset, including (a) reference, (b) simulated noisy SAR images, (c) real SAR images, and (d) CCMs.

We use supervised learning with a learning rate set to $1e^{-4}$. The batch size is 24, and a total of 100 epochs are trained. Every 40 epochs, the learning rate is reduced by a factor of 1/10. We utilize the adaptive moment estimation (ADAM) [57] optimizer and implement the model using the PyTorch framework.

A. Dataset

Due to the lack of widely used datasets for denoising multitemporal SAR images, we have created a new dataset specifically for training denoising networks. This dataset comprises 300 subsets, each containing a real SAR image along with its corresponding noise-free reference image, simulated noisy image, and CCM, as shown in Fig. 6. These data are all single-channel images with a size of 400 × 400, and the length of the time series is 5.

The reference images are obtained through stacked multilook processing, while the noisy images are generated by adding independently distributed speckle noise to the reference images. The CCM is computed using (5). This dataset represents the first publicly available resource for training denoising networks specifically for multitemporal SAR images, and it incorporates valuable physical priors.

B. Simulated Experiments

The purpose of this experiment is to validate our model's superior performance compared with other methods. The input image sizes are all 400 × 400, except for SAR-OCN, which requires 384 × 384. The length of the time series varies. We evaluate the model's performance using three commonly

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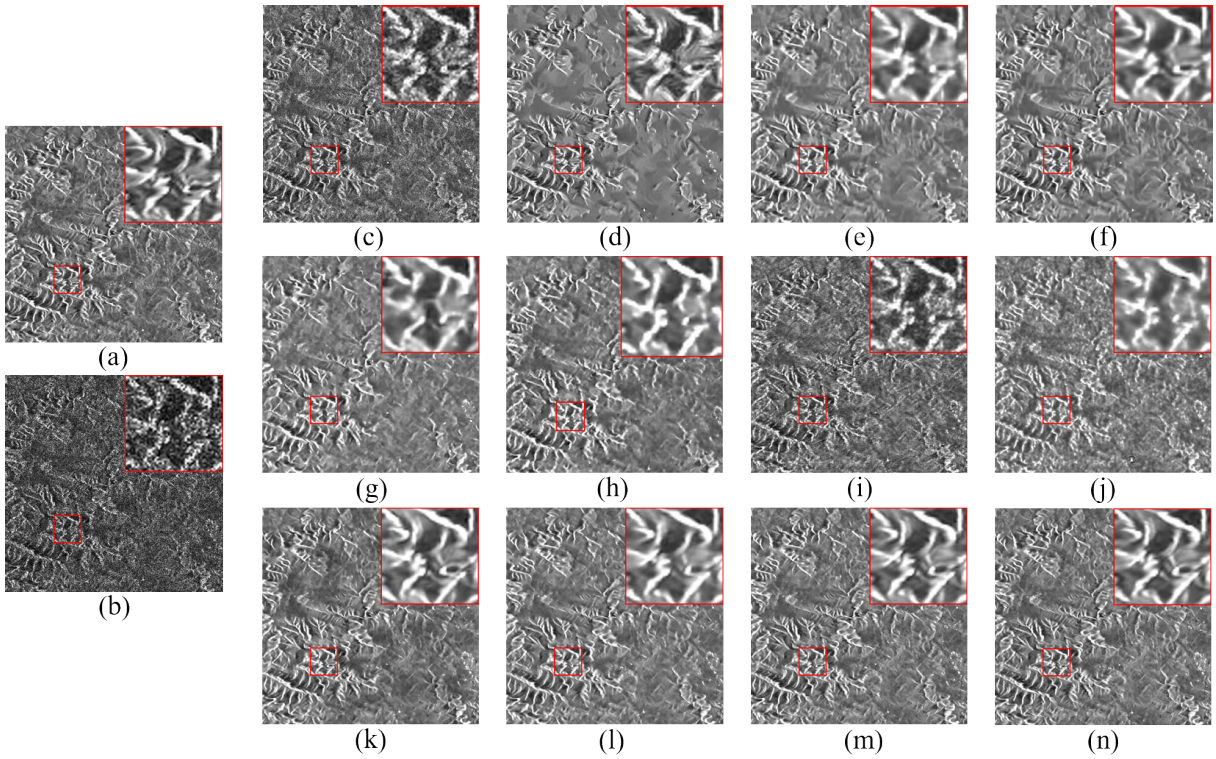


Fig. 7. Simulation experiment results in a mountain area. (a) Reference. (b) Speckle $L = 1$. (c) UTA ($M = 5$). (d) 2S-PPB ($M = 5$). (e) MSAR-BM3D ($M = 4$). (f) RABASAR ($M = 5$). (g) MONet ($M = 1$). (h) SAR-OCN ($M = 1$). (i) SAR2SAR ($M = 2$). (j) NR-SAR ($M = 2$). (k) MSAR-CNN ($M = 5$). (l) Proposed ($M = 2$). (m) Proposed ($M = 4$). (n) Proposed ($M = 5$).

used metrics: peak signal-to-noise ratio (PSNR) [58], structural similarity index (SSIM) [59], and erreur relative globale adimensionnelle de synthese (ERGAS) [60]. These measures are defined as follows:

$$\mathrm{PSNR} = 10\lg\left(\frac{\mathrm{MAX}^2}{\mathrm{mse}\left(X,\hat{X}\right)}\right) \tag{18}$$

$$\mathrm{SSIM} = \frac{\left(2\mu_{\hat{X}}\mu_X + C_1\right)\left(2\sigma_{\mu_{\hat{X}}\mu_X} + C_2\right)}{\left(\mu_{\hat{X}}^2 + \mu_X^2 + C_1\right)\left(\sigma_{\mu_{\hat{X}}}^2 + \sigma_{\mu_X}^2 + C_2\right)} \tag{19}$$

$$\mathrm{ERGAS} = 100\sqrt{\frac{1}{M}\left(\frac{\sqrt{\mathrm{mse}\left(\hat{X},X\right)}}{\mu}\right)^2}. \tag{20}$$

In (18), mse is the mean-squared error of X and $\hat{X}$, and MAX is the maximum pixel value of the image. In (19), μ denotes the means, and σ represents the standard deviation or cross covariance, respectively. C_1 and C_2 are constants. In (20), M represents the temporal number of images. μ denotes the average brightness. As can be seen from the definitions, a higher PSNR, an SSIM closer to 1, and a lower ERGAS indicate better denoising performance by the model.

To visually demonstrate the denoising performance of our model, we selected three images from the test dataset, corresponding to mountainous, urban, and flat regions. In mountainous regions, the adaptive noise estimation and texture preservation capabilities of noise reduction methods are visually evident. In urban areas with complex buildings and numerous corner reflectors, various methods' ability to preserve details becomes apparent. In uniform regions, where detail information is scarce, it is easier to visually judge whether noise has been completely removed. The denoising results for these three images are shown in Figs. 7–9. They display the visual outcomes of our proposed method and other comparative approaches. In these three figures, (l), (m), and (n) indicate that the proposed method is trained and tested on M temporal images, while the corresponding size of the time series for other methods has been described earlier in the text.

From a visual qualitative perspective, UTA exhibits significant noise residue. The 2S-PPB algorithm, while effective in denoising, tends to oversmooth images, resulting in the loss of fine details. Among model-based algorithms, MSAR-BM3D and RABASAR achieve the best visual results. Deep learning algorithms, leveraging their excellent nonlinear modeling capabilities, demonstrate good denoising performance in certain scenarios. Compared with other methods, our proposed approach exhibits less noise residue and enhanced texture details. Even with less temporal redundancy, it can achieve more refined results compared with traditional methods.

In the urban region (see Fig. 8), our method accurately preserves the texture of artificial structures, such as high rises in the city center and roads, effectively restoring the scattering structure around buildings (as indicated by the red box). With only two temporal images, the building outlines on both sides of the river are well restored. As the size of the time series increases, the bare ground textures around buildings are also depicted, demonstrating the preservation of fine details. In the mountain area (see Fig. 7), our proposed method maintains the original image resolution even under strong noise influence. It performs well in reconstructing darker and lower contrast

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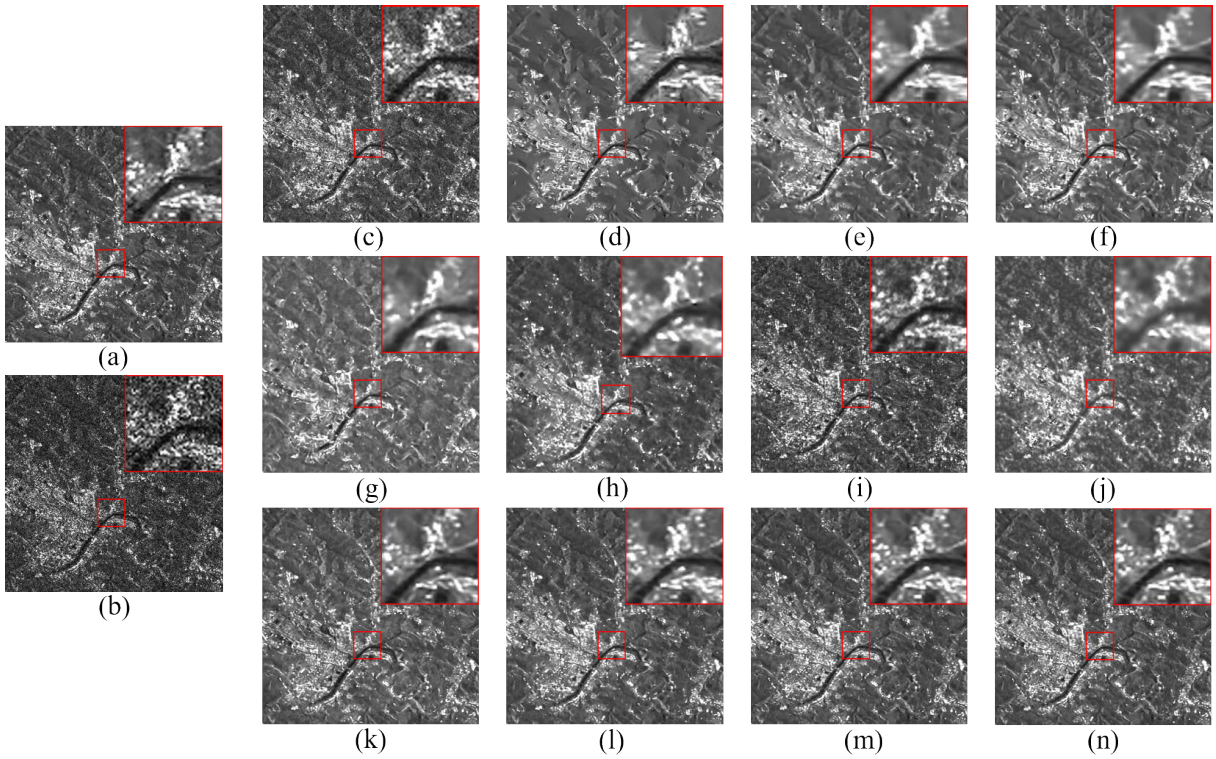


Fig. 8. Simulation experiment results in an urban area. (a) Reference. (b) Speckle $L = 1$. (c) UTA ($M = 5$). (d) 2S-PPB ($M = 5$). (e) MSAR-BM3D ($M = 4$). (f) RABASAR ($M = 5$). (g) MONet ($M = 1$). (h) SAR-OCN ($M = 1$). (i) SAR2SAR ($M = 2$). (j) NR-SAR ($M = 2$). (k) MSAR-CNN ($M = 5$). (l) Proposed ($M = 2$). (m) Proposed ($M = 4$). (n) Proposed ($M = 5$).

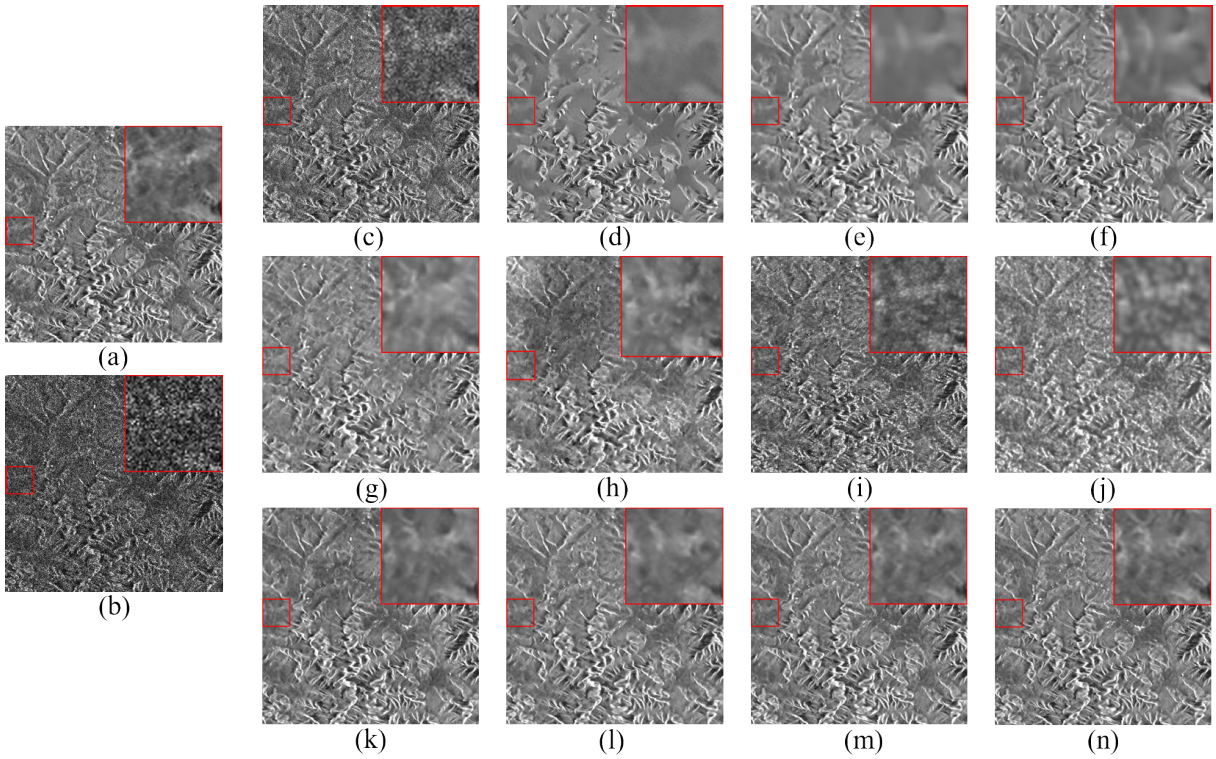


Fig. 9. Simulation experiment results in a flat area. (a) Reference. (b) Speckle $L = 1$. (c) UTA ($M = 5$). (d) 2S-PPB ($M = 5$). (e) MSAR-BM3D ($M = 4$). (f) RABASAR ($M = 5$). (g) MONet ($M = 1$). (h) SAR-OCN ($M = 1$). (i) SAR2SAR ($M = 2$). (j) NR-SAR ($M = 2$). (k) MSAR-CNN ($M = 5$). (l) Proposed ($M = 2$). (m) Proposed ($M = 4$). (n) Proposed ($M = 5$).

regions, capturing intricate textures unique to mountain farmland. In contrast, other methods, particularly MSAR-BM3D and RABASAR, tend to be overly smooth, while algorithms, such as MSAR-CNN and SAR-OCN, introduce artifacts into

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TABLE I
QUANTITATIVE INDICATORS OF THE THREE IMAGES

Sence	Evaluation	Noisy	MONet	SAR -OCN	SAR2SAR	NR -SAR	Proposed	MSAR -BM3D	Proposed	UTA	2S -PPB	RABASAR	MSAR -CNN	Proposed
			M=1	M=1	M=2	M=2	M=2	M=4	M=4	M=5	M=5	M=5	M=5	M=5
Mountain	PSNR	32.72	37.29	33.52	33.11	37.64	39.60	34.85	**41.58**	34.30	34.75	35.23	39.76	41.22
	SSIM	0.643	0.918	0.854	0.651	0.923	0.947	0.810	**0.968**	0.852	0.875	0.882	0.949	0.966
	ERGAS	0.262	0.201	0.249	0.256	0.197	0.176	0.231	**0.157**	0.239	0.233	0.226	0.174	0.160
Urban	PSNR	25.06	26.67	29.24	25.02	27.38	32.14	25.55	33.03	27.35	27.36	28.93	30.77	**33.11**
	SSIM	0.419	0.781	0.867	0.413	0.755	0.927	0.522	0.940	0.719	0.797	0.852	0.909	**0.945**
	ERGAS	0.294	0.270	0.228	0.295	0.257	0.196	0.286	0.186	0.258	0.258	0.235	0.212	**0.185**
Uniform	PSNR	32.78	35.08	32.97	35.35	37.48	40.26	38.54	40.97	36.49	38.32	39.74	39.94	**41.21**
	SSIM	0.690	0.874	0.838	0.850	0.914	0.954	0.938	0.961	0.880	0.935	0.949	0.952	**0.965**
	ERGAS	0.241	0.211	0.242	0.208	0.184	0.157	0.173	0.150	0.195	0.175	0.161	0.160	**0.148**

the image. Our proposed method does not introduce artifacts in the flat area (see Fig. 9) and achieves the best preservation of details. Although the denoised results obtained by training our proposed model on two temporal images do not capture as much detail as those trained on four or five, it achieves the best visual effect compared with contrast methods when the number of temporal images is the same. Compared with single-temporal methods, such as SAR-OCN, the additional temporal information utilized by the network in this study helps preserve spatial details and reduce blurring around sharp structures.

The quantitative indicators of these three SAR images are shown in Table I. In Table I, M represents the length of the time series. The optimal results are bolded, and the second-best results are indicated with an underscore.

In Table I, we arrange the methods that handle the same number of temporal images in adjacent positions to compare their evaluation metrics under the same input. When faced with strong noise, the proposed method demonstrates the most effective denoising across all three scenarios. In addition, due to the lack of spatiotemporal information, deep learning methods for single-temporal data processing perform less favorably in some scenarios compared with traditional multitemporal denoising methods. For example, in homogeneous and urban areas, MONet is outperformed by RABASAR in all three metrics, and in visual images, MONet results in more loss of detail compared with RABASAR. SAR2SAR, a classic multistage unsupervised network for processing multitemporal SAR images, is employed in many studies for comparative experiments. Moreover, the noise suppression level achieved in the actual denoising process is acceptable, so we also include SAR2SAR in the comparative experiments. In every size of the time series, the proposed method achieves the best quantitative evaluation. Furthermore, the proposed method obtains both the best and second-best quantitative evaluations, differing only in the number of temporal images. Apart from the proposed method, MSAR-CNN performs the best, largely due to its thorough integration of multitemporal features. Methods that are prone to causing loss of detail perform the worst in homogeneous areas, such as RABASAR and SAR-OCN, as they remove the fine texture of homogeneous areas to eliminate noise, resulting in a decline in quantitative metrics.

In addition, we separately analyzed the performance of different methods on the test dataset under varying noise intensities. The results are shown in Table II, where the values represent the averages of the metrics on the test images. Similarly, the best results are highlighted in bold, and the second-best results are indicated with an underscore.

From Table II, it can be observed that our method achieves the best denoising results across all levels of noise interference. In all temporal images, the proposed method also reflects the best quantitative evaluation, demonstrating the superior performance of the denoiser. In addition, we can draw several conclusions. First, as noise intensity decreases, the evaluation metrics for all denoised images gradually improve. Lower noise levels are generally more favorable for all denoising methods. Furthermore, data-driven deep learning methods perform well in denoising, likely due to their excellent nonlinear modeling capabilities. However, certain model-based algorithms that handle multitemporal data, such as RABASAR, outperform deep learning methods that operate solely on single-temporal images (e.g., SAR-OCN). Finally, when using deep learning methods to handle different noise intensities, each method employs only one model. This tests the model's generalization performance. Our proposed approach effectively handles various noise levels using a single model, demonstrating robust capabilities.

C. Real Experiments

To validate the effectiveness of the proposed method in practical applications, we evaluated filtering performance using real multitemporal SAR data. The data were sourced from the Sentinel-1 satellite, acquired in interferometric wide imaging mode and in single-look complex format. Due to interference from SAR imaging characteristics and internal sensor system factors, the original images consistently exhibited noticeable noise pollution. We selected a region in Shanxi, China, to compute coherence coefficients and build a dataset for the experiment. Both the CCM and SAR images are taken as input. We show three images in Figs. 10–12 for visual evaluation.

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TABLE II
AVERAGE PERFORMANCE MEASURES OF SIMULATED DENOISING RESULTS

Looks	Evaluation	Noisy	MONet -OCN	SAR2SAR	NR -SAR	Proposed	MSAR -BM3D	Proposed	UTA -PPB	RABASAR	MSAR -CNN	Proposed
			M=1	M=1	M=2	M=2	M=2	M=4	M=4	M=5	M=5	M=5	M=5
1	PSNR	30.36	32.48	33.13	31.74	33.77	36.75	32.38	37.59	33.24	33.71	34.38	36.01	**38.18**
	SSIM	0.585	0.858	0.871	0.697	0.871	0.936	0.736	0.946	0.817	0.870	0.881	0.929	**0.954**
	ERGAS	0.283	0.252	0.242	0.263	0.234	0.196	0.256	0.187	0.242	0.237	0.227	0.206	**0.181**
2	PSNR	31.60	33.39	35.65	33.32	35.44	38.30	33.83	39.28	34.85	33.98	35.59	37.49	**39.46**
	SSIM	0.684	0.877	0.921	0.792	0.907	0.954	0.810	0.964	0.880	0.886	0.900	0.948	**0.965**
	ERGAS	0.264	0.240	0.209	0.241	0.212	0.179	0.235	0.170	0.221	0.233	0.212	0.189	**0.168**
4	PSNR	33.43	33.84	37.59	34.75	36.46	39.69	36.01	40.40	36.91	34.58	36.85	39.47	**41.01**
	SSIM	0.797	0.868	0.946	0.863	0.927	0.966	0.883	0.969	0.933	0.911	0.921	0.966	**0.975**
	ERGAS	0.238	0.235	0.187	0.223	0.200	0.165	0.209	0.159	0.197	0.225	0.198	0.168	**0.154**
8	PSNR	35.52	34.10	38.60	34.95	37.30	40.94	37.63	41.37	38.04	34.73	37.59	40.51	**41.56**
	SSIM	0.876	0.848	0.955	0.872	0.937	0.974	0.921	**0.976**	0.951	0.923	0.934	0.972	**0.976**
	ERGAS	0.211	0.234	0.176	0.219	0.190	0.154	0.189	0.150	0.185	0.223	0.189	0.158	**0.149**

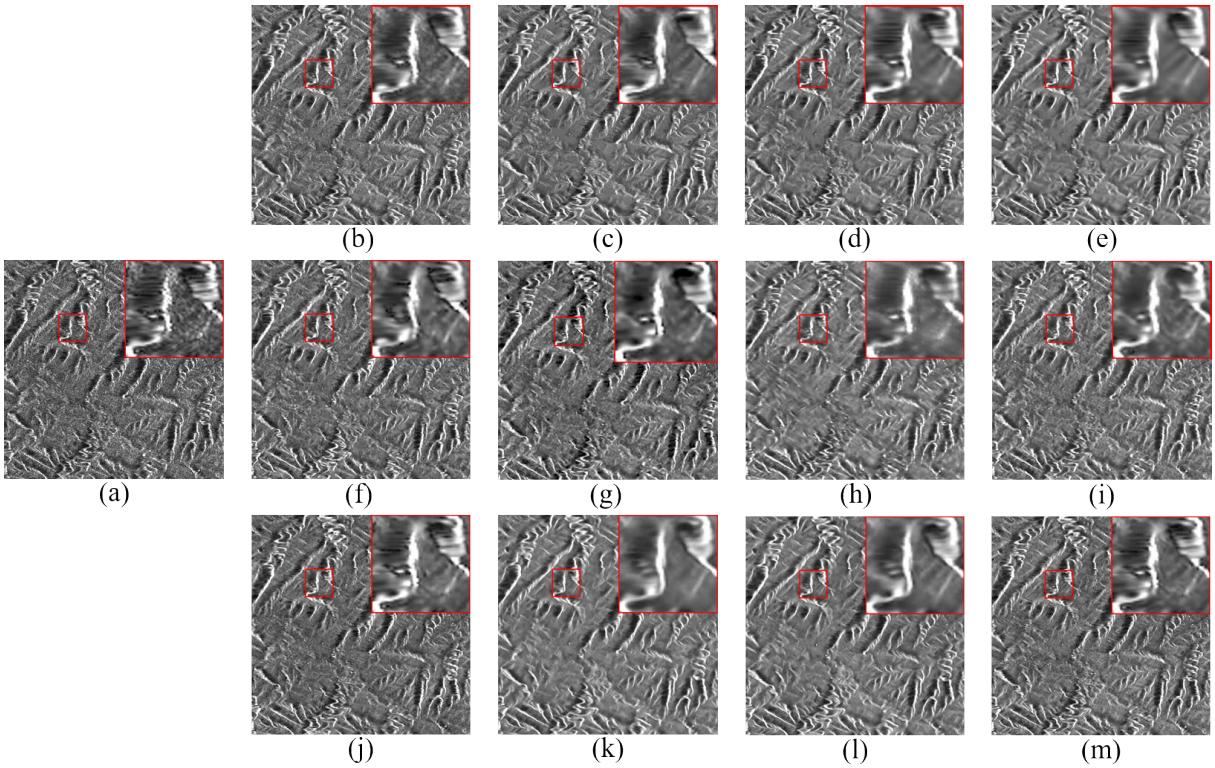


Fig. 10. Real experiment results in a mountain area. (a) Realistic noisy image. (b) UTA ($M = 5$). (c) 2S-PPB ($M = 5$). (d) MSAR-BM3D ($M = 4$). (e) RABASAR ($M = 5$). (f) MONet ($M = 1$). (g) SAR-OCN ($M = 1$). (h) SAR2SAR ($M = 2$). (i) NR-SAR ($M = 2$). (j) MSAR-CNN ($M = 5$). (k) Proposed ($M = 2$). (l) Proposed ($M = 4$). (m) Proposed ($M = 5$).

In these images, the land cover types are diverse, including urban, flat, and mountain ranges. Since there are no truly noise-free SAR images available for reference, our judgment of denoising results is based on subjective experience. We consider a method superior if its denoised output effectively restores image details and textures while maintaining local smoothness. Upon examining the denoising results, we observe that some methods fail to completely remove noise, while others sacrifice image details, particularly noticeable textures. Single-temporal methods eliminate noise but blur the image, resulting in irregular patches. Multitemporal methods perform slightly better but still exhibit some issues. MSAR-BM3D produces pronounced artifacts in certain mountainous regions, as highlighted in the boxed area. In regions with high information density, such as Fig. 11(i), NR-SAR fails to reconstruct the image based on the actual structure, resulting

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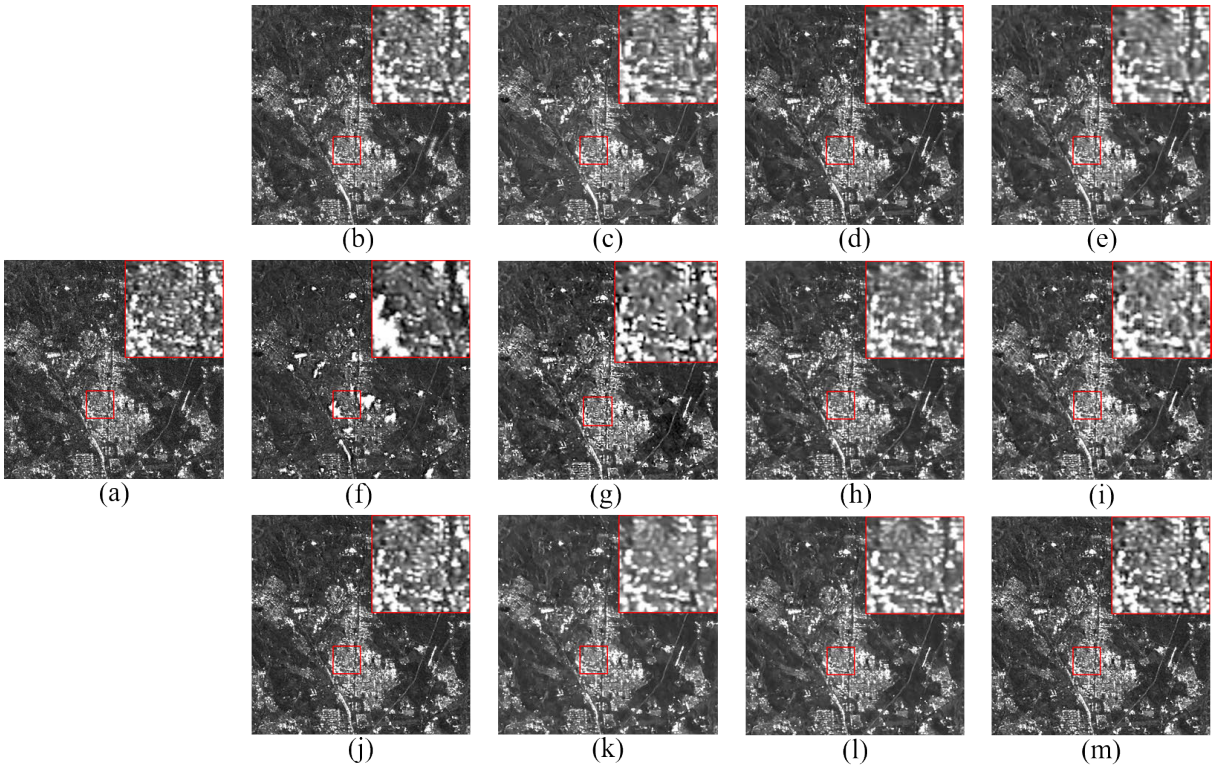


Fig. 11. Real experiment results in an urban area. (a) Realistic noisy image. (b) UTA ($M = 5$). (c) 2S-PPB ($M = 5$). (d) MSAR-BM3D ($M = 4$). (e) RABASAR ($M = 5$). (f) MONet ($M = 1$). (g) SAR-OCN ($M = 1$). (h) SAR2SAR ($M = 2$). (i) NR-SAR ($M = 2$). (j) MSAR-CNN ($M = 5$). (k) Proposed ($M = 2$). (l) Proposed ($M = 4$). (m) Proposed ($M = 5$).

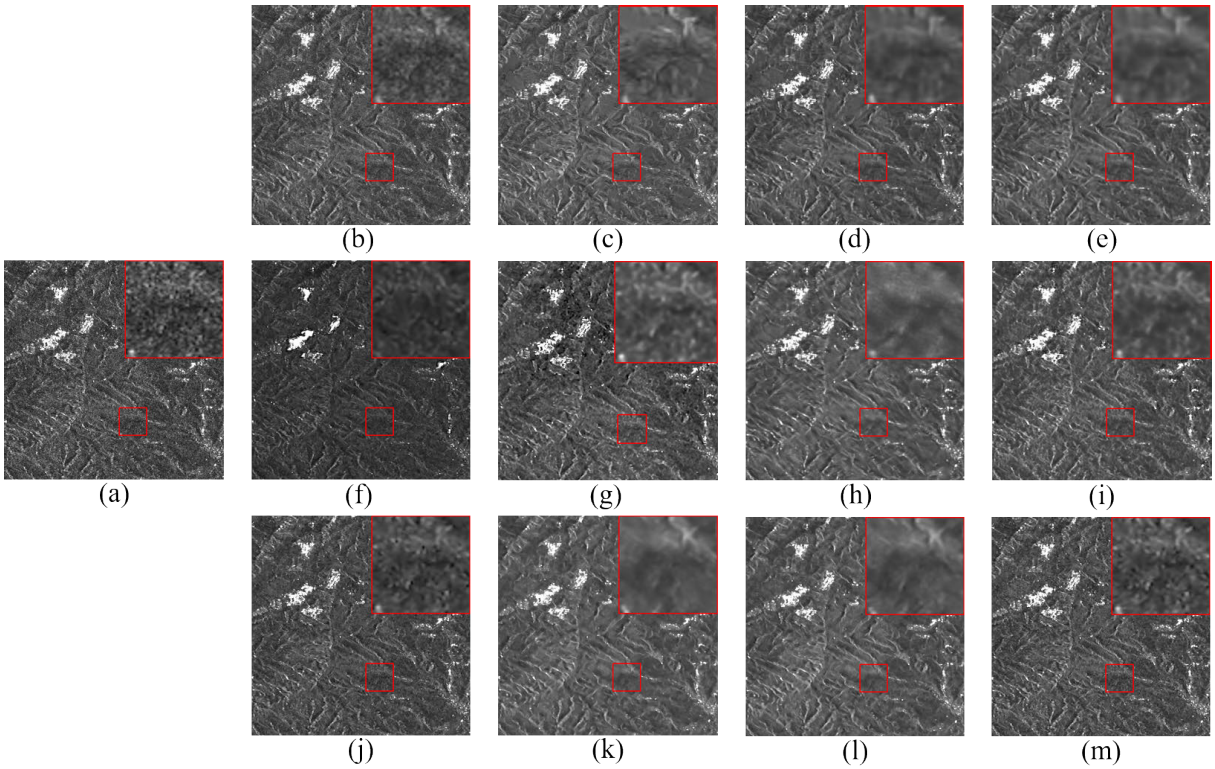


Fig. 12. Real experiment results in a flat area. (a) Realistic noisy image. (b) UTA ($M = 5$). (c) 2S-PPB ($M = 5$). (d) MSAR-BM3D ($M = 4$). (e) RABASAR ($M = 5$). (f) MONet ($M = 1$). (g) SAR-OCN ($M = 1$). (h) SAR2SAR ($M = 2$). (i) NR-SAR ($M = 2$). (j) MSAR-CNN ($M = 5$). (k) Proposed ($M = 2$). (l) Proposed ($M = 4$). (m) Proposed ($M = 5$).

in the appearance of regularly arranged false point targets. RABASAR's denoising results suffer from oversmoothing, leading to significant information loss. Compared with other multitemporal deep learning methods, our proposed approach

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TABLE III
AVERAGE PERFORMANCE MEASURES OF REAL DENOISING RESULTS

Evaluation	MONet	SAR-OCN	SAR2SAR	NR-SAR	Proposed	MSAR BM3D	Proposed	UTA	2S-PPB	RABASAR	MSAR CNN	Proposed
	M=1	M=1	M=2	M=2	M=2	M=4	M=4	M=5	M=5	M=5	M=5	M=5
ENL	**6.691**	3.223	5.282	3.985	3.086	3.192	3.078	3.094	4.353	4.849	2.960	3.351
EPD-ROA	0.463	0.600	0.482	**0.642**	0.588	0.558	0.549	0.556	0.617	0.585	0.581	0.623

preserves texture orientation and the shapes of small bright targets, with minimal artifacts. Our proposed method deeply understands and distinguishes between noise and dark spots caused by ground features, removing noise while preserving structural information and effective textures for the dark areas appearing in the images (see Fig. 12). In addition, within all the time-series lengths, the proposed method restores the image reflectance to match actual observed values best. With the increase in the number of temporal images, edges become clearer, and low-contrast structures are better preserved.

In this study, we quantitatively evaluated the performance using two key indicators: equivalent number of looks (ENL) [61] and edge preservation degree based on ratio of the average (EPD-ROA) [62]. They are defined as (21) and (22). ENL reflects the intensity of multiplicative noise present in SAR images, where higher values correspond to smoother images. EPD-ROA represents the degree of edge preservation in an image. The closer its value is to 1, the better the filtering algorithm performs in maintaining edge details. It is important to note that these evaluation metrics serve as a reference and do not carry definitive significance. ENL and EPD-ROA are expressed as follows:

$$\text{ENL} = \left(\frac{\mu}{\sigma}\right)^2 \tag{21}$$

$$\text{EPD-ROA} = \frac{\sum_{i=1}^{m} |u_{D_1}(i)/u_{D_2}(i)|}{\sum_{i=1}^{m} |u_{O_1}(i)/u_{O_2}(i)|}. \tag{22}$$

In (21), μ is the expectation of the image, and σ denotes variance. In (22), m is the number of pixels in the image. $u_{D_1}(i)$ and $u_{D_2}(i)$, respectively, represent the intensity values of two adjacent pixels in a certain direction in the denoised image. Similarly, $u_{O_1}(i)$ and $u_{O_2}(i)$ represent the intensity values of two adjacent pixels in the original image in another direction.

We calculated these two quantitative metrics based on the denoising results of real multitemporal SAR images from our test dataset across various comparative methods. The results are summarized in Table III. Although the proposed method did not achieve the best results in these metrics, it remains at an acceptable level overall. In the EPD-ROA, our proposed method received the second-best evaluation. The method that achieved the best metrics exhibited structural overflow and artifacts in the visual results, which is related to the singular nature of these metric evaluations. Images with excessively high ENL tend to be overly smooth, resulting in poorer outcomes. By combining these quantitative metric evaluations with visual assessments, it can be observed that our method achieved the best denoising effect.

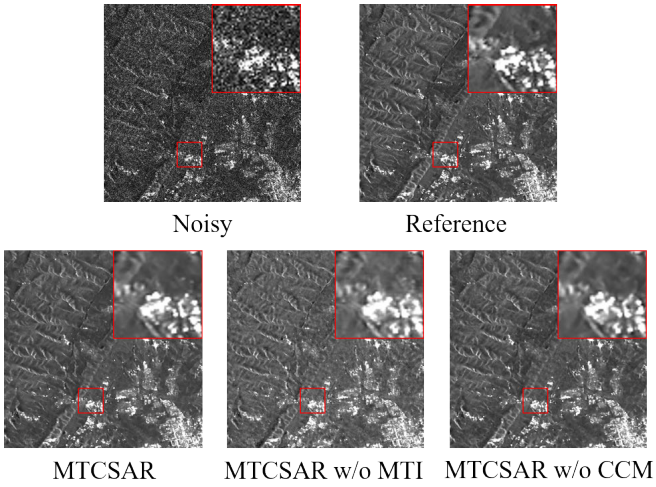


Fig. 13. Visual results of ablation experiments. The complete network achieved the best performance.

D. Ablation Experiment

The primary architecture of this network comprises three main modules: the image global MTI module, the fusion module, and the CCM prior feature extraction module. To investigate the roles and significance of these modules within the network, we conducted ablation experiments. Specifically, we removed each of these modules separately and retrained the model. We then compared the denoising performance of these models on a simulated multitemporal image test dataset. Our study includes two sets of ablation experiments: one targeting the MTI and the other focusing on the CFE module. Through these experiments, we aim to explore the necessity of the proposed MTI and the injection of prior information.

In Fig. 13, we present the recovery results of the complete network and the ablated network (denoted as "w/o") in an urban area under the influence of gamma noise. From a visual perspective, when the network lacks the integration of temporal redundancy information through self-attention mechanisms, it generates significant artifacts in both horizontal and vertical directions in flat regions (as highlighted in the enlarged portion of the image in MTCSAR w/o MTI). In addition, in scenes with high-contrast targets, all types of targets are adversely affected. The network without CCM, due to the absence of prior information injection, fails to adequately handle temporal information, resulting in oversmoothed images and partial loss of texture.

The quantitative metrics, summarized in Table IV, reveal that even after ablation, the denoising performance of the

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TABLE IV
AVERAGE PERFORMANCE MEASURES OF ABLATION DENOISING RESULTS

Looks	Evaluation	Noisy	Proposed	Proposed w/o MTI	Proposed w/o CCM
1	PSNR	30.36	**38.18**	36.09	36.33
	SSIM	0.585	**0.954**	0.925	0.931
	ERGAS	0.283	**0.181**	0.205	0.202
2	PSNR	31.60	**39.46**	36.98	38.03
	SSIM	0.684	**0.965**	0.933	0.952
	ERGAS	0.264	**0.168**	0.194	0.183
4	PSNR	33.43	**41.01**	38.39	39.86
	SSIM	0.797	**0.975**	0.947	0.968
	ERGAS	0.238	**0.154**	0.179	0.165
8	PSNR	35.52	**41.56**	38.24	40.80
	SSIM	0.876	**0.976**	0.942	0.972
	ERGAS	0.211	**0.149**	0.180	0.156

network remains relatively high for noisy images. However, both ablated networks exhibit inferior denoising capabilities compared with the complete network. This observation underscores the beneficial roles played by the self-attention mechanism across multitemporal and the CCM features based on physical properties. When comparing the two ablated networks, denoising indicators favor the network equipped with MTI over the one with CCM. Although coherence coefficients provide prior information about neighboring temporal images relative to a reference, the pointwise computation limitation restricts CCM's assistance primarily to the time dimension. In contrast, MTI constructs spatiotemporal windows for self-attention, enabling deep information exploration across both time and space, thus contributing more significantly to the network's performance.

E. Performance of Changes in Multitemporal SAR Images

Temporal leakage poses a significant challenge in multitemporal image denoising and super-resolution. This phenomenon occurs when erroneous information from other dates contaminates the reference date, thereby questioning the authenticity of the restored reference. Our network effectively addresses this issue by enhancing the credibility of denoising results for the reference date. To evaluate the performance of various methods on multitemporal SAR images, we selected two dates with a substantial interval from the same time series as reference dates. We then compared the results obtained by other methods for these two dates with those obtained by our proposed method, assessing their ability to manage change information contamination in multitemporal SAR images.

We present the results of various multitemporal methods for handling change information in Fig. 14. In the two subfigures corresponding to each method, the top figure represents the date named Date1, and the bottom figure represents another date, Date2. To more clearly display the handling of change information, areas with significant changes are marked with yellow ellipses within the red box. As shown in Fig. 14(a), from Date1 to Date2, the larger elliptical area exhibits noticeable bright spot targets, while the smaller elliptical area shows dimming of some point targets. Fig. 14(c)–(f) is significantly affected by false information contamination from other dates, showing noticeable attenuation. This indicates that UTA, MSAR-BM3D, RABASAR, and 2S-PPB cannot effectively prevent interference from changing information. Among the data-based methods, Fig. 14(g) shows overflow in the box area, while Fig. 14(h) exhibits regularly distributed hidden artifacts. Compared with Fig. 14(i), which performed suboptimally in visual qualitative results, our proposed method identified subtle changes in the smaller elliptical area and exceled in texture restoration.

To objectively evaluate the anti-interference capability of various methods against change information, we propose a quantitative evaluation metric called multitemporal PSNR (MTPSNR). This metric transforms the PSNR values calculated for the denoised results ($\hat{X}_1$ and $\hat{X}_2$) of two reference dates (X_1 and X_2), assessing the preservation of the reference date features (X_1 and X_2) in the denoised images ($\hat{X}_1$ and $\hat{X}_2$) under the contamination of change information from other dates in the time series. This metric is defined as follows:

$$\mathrm{MTPSNR} = \sqrt{\frac{\mathrm{PSNR}\left(X_1, \hat{X}_1\right)^2 + \mathrm{PSNR}\left(X_2, \hat{X}_2\right)^2}{2}} \tag{23}$$

where PSNR refers to the peak signal-to-noise ratio between two inputs, calculated using (18). A higher MTPSNR value indicates that the restoration of the two reference dates is closer to the original reference date images, making them less susceptible to contamination from changing information on other dates.

As shown in Table V, our method achieved the highest MTPSNR rating, indicating it is closest to the target dates and most effectively avoids contamination from changing information on other dates. MSAR-CNN obtained the second-best results, consistent with the conclusions from visual qualitative evaluations. Other methods received unsatisfactory quantitative evaluations, indicating a lower capability to handle false information.

V. DISCUSSION

A. Impact of MTIB on Performance

The MTI module of MTCSAR consists of multiple MTIB stacked together, except for the first convolutional layer. This structure allows for the reuse of the stacked portions. As explained in Section IV-D, the impact of spatiotemporal self-attention information extraction mechanism significantly influences the network's performance. Therefore, based on varying quantities of stacked MTIB, we designed three networks with different parameter sizes but identical overall structures: MTCSAR-Tiny, MTCSAR-Base, and MTCSAR-Large. Their MTIB stack numbers are 1, 2, and 4, respectively. The MTCSAR-Base corresponds to the proposed network discussed in Section IV. Their parameter counts, average denoising metrics on the test image dataset, and test times are summarized in Table VI.

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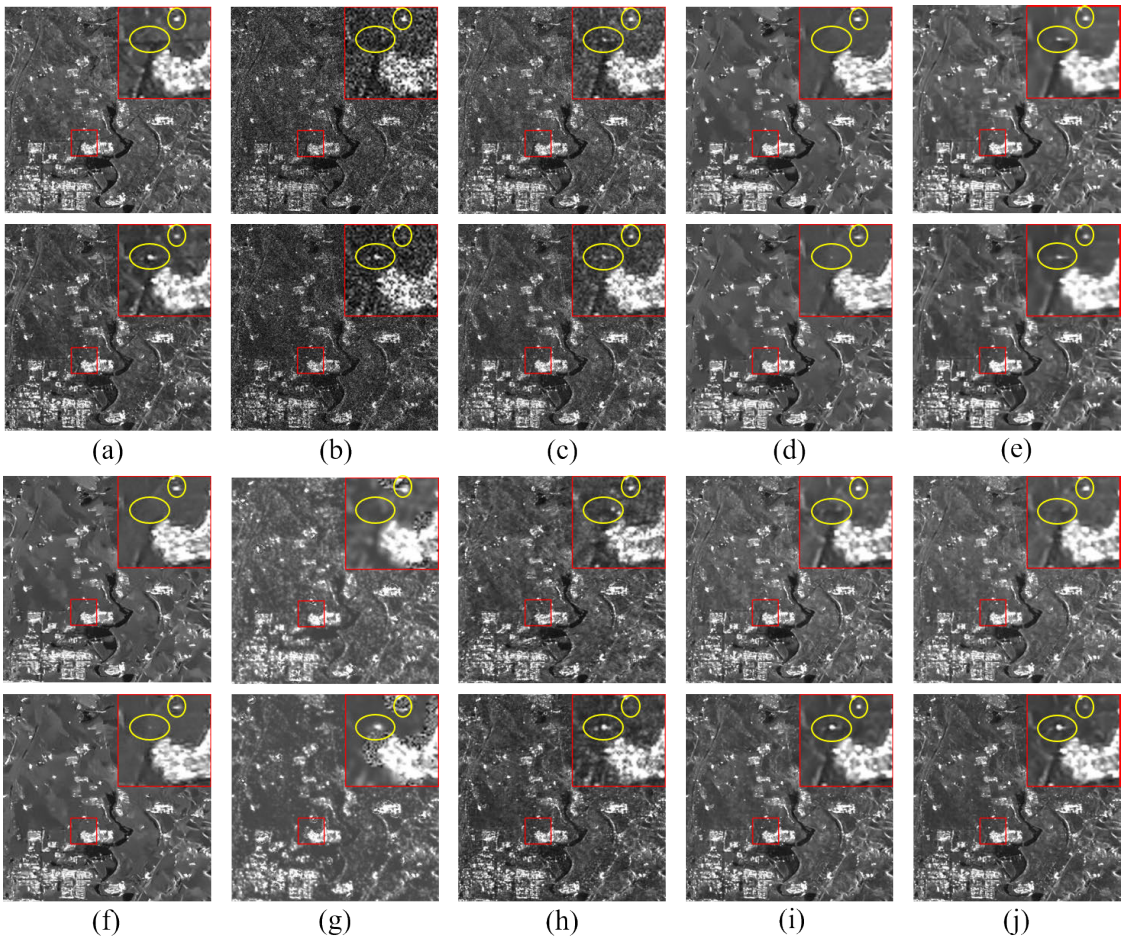


Fig. 14. Denoising results targeting two dates. They are located in the top and bottom rows, respectively. (a) Reference. (b) Speckle $L = 1$. (c) UTA. (d) MSAR-BM3D. (e) RABASAR. (f) 2S-PPB. (g) NR-SAR. (h) SAR2SAR. (i) MSAR-CNN. (j) Proposed.

TABLE V
MTPSNR OF DENOISING RESULTS

Evaluation	UTA	MSAR-BM3D	RABASAR	2S-PPB	NR-SAR	SAR2SAR	MSAR-CNN	Proposed
MTPSNR	35.54	34.65	35.99	34.05	34.20	32.92	38.25	**39.04**

TABLE VI
AVERAGE PERFORMANCE MEASURES OF DIFFERENT MTCSARS

MTCSAR Version	Denoising performance			Times (s/item)	Parameters (M)	FLOPs (G)
	PNSR	SSIM	ERGAS			
MTCSAR-T	40.038	0.9676	0.1630	0.3191	1.8384	10.880
MTCSAR-B	40.051	0.9677	0.1630	0.5515	2.4108	15.183
MTCSAR-L	40.094	0.9678	0.1626	0.7788	3.5555	23.789

As the number of MTIBs increases, the network parameters grow, and the runtime gradually lengthens. Meanwhile, the denoising performance of the network improves very slowly, as MTIBs have a sophisticated structure that can fully extract image information with just a few layers. Considering computational complexity and network performance, we generally find that MTCSAR-Base yields the best results.

B. Impact of the Length of Time Series

Theoretically, adding more input dates should improve the quality of the predicted denoised image. However, any changes in the time series could affect the performance of the denoising process. In Section IV-D, we have already demonstrated the effectiveness of the proposed method in managing change information. Here, we will quantitatively assess the network's performance when additional dates are included.

We consider the scenario where the network processes a single SAR image. Notably, a single image cannot compute CCM, so we slightly modified the network by removing the CFE and the CCM attention module, forming MTCSAR-Single. After modification, the network input is a single SAR intensity image, and the output is the denoised result. The network for processing other time-series lengths remains the same, as described in Section IV. Fig. 15 shows the line graph of PSNR calculated with the increasing number of additional dates. Our network can filter single SAR images and performs better than other methods that process a single date. In the test dataset, the average PSNR value of MTCSAR-Single is higher than that of MONet (33.45 dB) and SAR-OCN (36.24 dB). PSNR and SSIM allow us to measure image restoration quality. Our network achieves the

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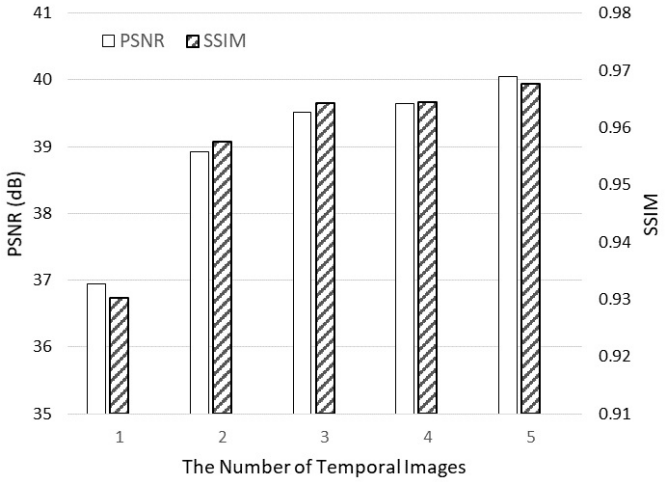


Fig. 15. PSNR with the increasing number of temporal images.

most significant improvement when the first additional date is present. Subsequently, as the number of additional dates increases, the network's improvement shows marginal effects, with diminishing returns.

VI. Conclusion

In this article, to address the issue of lacking temporal information for denoising SAR images, we leverage the redundancy of multitemporal data and combine it with coherence prior knowledge. We propose a novel network called MTCSAR, which fuses MTI with coherence prior for SAR image denoising. Within the network, an MTI module is introduced to extract long-term global features, while coherence priors are used to capture local features. The DAF module enhances information interaction and maintains spatiotemporal consistency. Various experiments demonstrate that the proposed network performs better in both simulated and real scenarios, accurately identifying interference caused by changes in neighboring temporal images. However, due to the adoption of a Transformer-based architecture, the network's computational complexity is high, even with the introduction of an anchor. In future work, we aim to enhance the network and reduce computational costs to achieve real-time performance. In addition, we plan to extend the application of denoising preprocessing to tasks, such as SAR target inspection and recognition.

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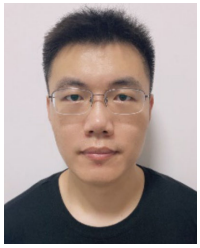
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